AI457 INTERNET OF THINGS & EDGE COMPUTING • COURSE PROJECT

IoT Smart Climate Monitoring & Automated AC Controller

End-to-End IoT Pipeline: Virtual Sensor Edge, Cloud Telemetry & Autonomous Climate Regulation

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WOKWI + PICO W

Edge Sensing & Logging

- Raspberry Pi Pico W with DHT22 sensor on GP0.
- Local CSV data logging and hardware LED alert.
- MicroPython firmware with 20s interval control.

THINGSPEAK CLOUD

Cloud IoT Telemetry

- Dual-variable time-series cloud repository.
- Field 1 (Temperature) & Field 2 (Humidity).
- RESTful Write and Read API integration.

LOCAL DASHBOARD

Smart AC Automation

- Modern, responsive web dashboard with Chart.js.
- Autonomous dual-variable climate decision engine.
- Energy-efficient cooling and humidity extraction.

PROJECT MOTIVATION & PROBLEM STATEMENT

Challenges in Modern Indoor Climate Regulation

COMFORT INEFFICIENCY

Static & Manual Thermostats

- Conventional ACs rely on rigid, manual setpoints.
- Fail to adapt to sudden ambient weather shifts.
- Frequent thermal discomfort & manual readjustments.
- Occupants left in overheated or excessively chilled rooms.

RESOURCE DRAIN

Severe Energy Waste

- Compressors cycle needlessly without real-time context.
- Temperature-only controls ignore humidity completely.
- Up to 35% of electricity is wasted on uncoordinated cooling.
- High utility bills and inflated residential carbon footprints.

DATA ISOLATION

Zero Remote Telemetry

- Traditional climate units operate in complete isolation.
- No historical telemetry logging or condition monitoring.
- Inability to audit energy trends or detect edge faults.
- Lack of unified web dashboard for real-time visibility.

The Core Objective

Build an autonomous closed-loop IoT system bridging edge sensing, cloud telemetry, and intelligent AC regulation.

Closed-Loop IoT

SYSTEM ARCHITECTURE & DATA FLOW

5-Stage Edge-to-Cloud Closed-Loop Pipeline

STAGE 1

Edge Sensing

- DHT22 digital sensor measures ambient Temp & Humidity.
- Single-wire digital pulse train connected to Pico W GP0.

STAGE 2

Edge Processing

- MicroPython firmware enforces 20s sampling cadence.
- Local CSV persistence and hardware LED alert on GP1.

STAGE 3

Cloud Ingestion

- Pico W connects to Wi-Fi & calls ThingSpeak Write API.
- Transmits Field 1 (Temp) and Field 2 (Hum) over HTTP.

STAGE 4

Cloud Storage

- ThingSpeak records telemetry with UTC timestamps.
- Renders live web charts and persists historical records.

STAGE 5

Smart Actuation

- Local dashboard fetches feeds via ThingSpeak Read API.
- Executes autonomous Smart AC control decision matrix.

Seamless Closed-Loop Workflow

Continuous 20-second telemetry cycle from physical edge sensing to cloud repository and automated home climate decision.

< 20s Cycle Latency

EDGE HARDWARE SIMULATION

Virtual Hardware Prototyping on Wokwi

MICROCONTROLLER

Raspberry Pi Pico W

- RP2040 Dual-core ARM Cortex-M0+ running at 133 MHz.
- Infineon CYW43439 2.4 GHz 802.11n Wi-Fi subsystem.
- Executes lightweight MicroPython firmware with low latency.
- Authenticates with Wokwi-GUEST open virtual Wi-Fi network.

CLIMATE SENSOR

DHT22 Digital Sensor

- Precision capacitive humidity sensor & thermistor elements.
- Single-wire digital protocol connected directly to GP0.
- Temperature Range: -40°C to +80°C (±0.5°C accuracy).
- Humidity Range: 0% to 100% RH (±2% accuracy).

ACTUATOR & BUS

Status LED & Power Rail

- Visual alert LED wired to GP1 with current-limiting resistor.
- Hardware safety threshold: LED illuminates when Temp > 30°C.
- Regulated 3.3V power supply rail and common ground bus.
- Provides instantaneous edge-level physical alert feedback.

High Simulation Fidelity

Wokwi accurately emulates RP2040 peripheral timing, Wi-Fi networking stack, and digital sensor signal protocols.

Zero Hardware Wear

EDGE SOFTWARE & EMBEDDED LOGIC

MicroPython Firmware & Sensor Acquisition

CORE SETUP

Wi-Fi & Sensor Init

- Initializes network interface via network.WLAN(STA_IF).
- Non-blocking connection loop verifies IP lease assignment.
- Instantiates sensor driver via dht.DHT22(Pin(0)).
- Configures Pin(1, OUT) for local alert LED actuation.

RATE LIMITING

20s Cadence Control

- Executes sample collection across 30 sequential readings.
- time.sleep(20) enforces strict 20-second interval.
- Prevents self-heating distortion inside the DHT22 chamber.
- Complies with ThingSpeak cloud rate limits (>= 15s).

EDGE SAFETY

Local Threshold Alert

- Autonomous edge check: if temp > 30: led.value(1) else: 0.
- Operates with zero latency independent of cloud connection.
- Logs every measurement to persistent local CSV file.
- Transmits real-time packet via urequests HTTP client.

Edge Firmware Loop Execution

sensor.measure() → read temp/hum → local safety check → CSV append → ThingSpeak HTTP upload → sleep(20s).

160 Lines of Code

EDGE COMPUTING & DATA INTEGRITY

Local CSV Logging & On-Chip Analytics

LOCAL DATA RESILIENCE

Persistent Local CSV Logging (dht22_log.csv)

- Stores all readings on the Pico W flash memory in standard CSV format.
- Header schema: Reading, Time_s, Temperature_C, Humidity_%.
- Guarantees data retention even during Wi-Fi drops or network disconnects.
- Enables offline audits, edge verification, and batch dataset export.
- Appends verified data rows incrementally after each sensor poll.

LOCAL ANALYTICS

On-Chip Edge Statistical Computation

- Buffers readings inside memory lists (temps[], hums[]).
- Computes Temperature Summary: Minimum, Maximum, and Mean (°C).
- Computes Humidity Summary: Minimum, Maximum, and Mean (%).
- Outputs complete statistical report to Serial Monitor after 30 cycles.
- Demonstrates edge computing: raw filtering & aggregation before cloud.

Edge Computing Principle

Performs initial data validation, statistical aggregation, and safety threshold actuation directly at the edge.

Fault-Tolerant Edge

CLOUD INTEGRATION & REST APIS

ThingSpeak Cloud Ingestion & Write API

CLOUD REPOSITORY

ThingSpeak IoT Platform

- MATLAB-powered cloud repository built specifically for IoT.
- Provides persistent multi-channel time-series databases.
- Scalable ingestion engine supporting millions of devices.
- Seamless integration with web dashboards & analytics.

HTTP INGESTION

RESTful Write API Endpoint

- Pico W issues HTTP GET query using urequests library.
- Endpoint: https://api.thingspeak.com/update.
- Parameters: ?api_key=H79CEL...&field1=temp&field2=hum.
- Payload transmits dual-variable climate data in real-time.

ACK VERIFICATION

Cloud Ingestion Feedback

- Server authenticates Channel Write API Key on each hit.
- Returns incrementing integer entry ID (1, 2, 3... 30).
- Serial monitor logs: ThingSpeak Response: N on success.
- Confirms zero packet loss during continuous 10-minute trial.

Channel Field Mapping

Field 1 = Ambient Temperature (°C) • Field 2 = Relative Humidity (%) • Enforced 20s interval guarantees 100% delivery.

HTTP 200 OK

CLOUD TELEMETRY & PERSISTENCE

Cloud Telemetry Persistence & Dynamic Graphs

DATA STORAGE

Time-Series Repository

- Automatically stamps every payload with UTC timestamps.
- Maintains strict chronological sequence across all readings.
- Stores entries reliably in cloud database with zero ops.
- Enables longitudinal seasonal tracking & trend analytics.

LIVE VISUALIZATION

Dual Real-Time Graphs

- Generates real-time Field 1 graph for Ambient Temperature.
- Generates real-time Field 2 graph for Relative Humidity.
- Auto-scaling axes with customizable smoothing & timeframes.
- Verifies that cloud receives and plots sensor data correctly.

SECURITY MODEL

Access Control Governance

- Separates Write API Key (sensor only) from Read API Key (dashboard).
- Supports public dashboard sharing or private token auth.
- Prevents unauthorized data injection into sensor channel.
- Decouples data acquisition from client-side visualization.

Decoupled Architecture

The edge sensor and laptop dashboard remain completely decoupled—communicating exclusively through the cloud broker.

Zero Data Silos

CLIENT-SIDE WEB ARCHITECTURE

Modern Client-Side Web Dashboard Architecture

CLEAN MINIMAL UI

HTML5 & CSS3 Interface

- Modern, responsive layout inspired by data analytics portals.
- Subtle rounded cards, generous whitespace, and soft shadows.
- Color-coded design: Coral (Temp), Sky-Blue (Hum), Green (AC).
- Fully responsive across laptops, tablets, and desktop displays.

CLIENT-SIDE ENGINE

Asynchronous JS Client

- Pure client-side JavaScript architecture (script.js).
- Uses native async fetch() directly against ThingSpeak Read API.
- Zero backend required (no Flask/Node); runs directly in browser.
- Local persistence of Channel ID and API Key in localStorage.

INTERACTIVE CHARTS

Chart.js Data Engine

- Industry-standard HTML5 canvas charting engine.
- Renders dual separated historical trend charts with smooth curves.
- Hover tooltips with exact values and measurement timestamps.
- Skips full redraw animations on poll for flicker-free updates.

Clean Codebase Organization

index.html (Semantic UI Layout) • style.css (Clean Theme) • script.js (API Client & Decision Engine) • README.md.

100% Client-Side

CLOUD-TO-CLIENT INTEGRATION

ThingSpeak Read API & Synchronization Engine

DATA INGESTION

Read API REST Endpoint

- URL: https://api.thingspeak.com/channels/{ID}/feeds.json.
- Parameters: ?results=30&api_key={READ_API_KEY}.
- Fetches recent 30 entries matching Pico W logging batch.
- Retrieves complete JSON envelope with feed timestamps.

DATA VALIDATION

Data Sanitization & Parsing

- Extracts numeric values for field1 (Temp) and field2 (Hum).
- Filters out NaN or corrupt entries to protect visualizations.
- Parses ISO 8601 timestamps into local human-readable time.
- Feeds structured arrays to cards, charts, and decision engine.

REAL-TIME CADENCE

Synchronized Polling Timer

- Integrated 20-second countdown timer displayed in header (20s → 0s).
- Automatically triggers fresh data fetch when timer reaches zero.
- Matches the 20-second edge transmission cadence of the Pico W.
- Manual 'Refresh' button allows instant on-demand sync anytime.

In-Browser Configuration Drawer

Users can configure Channel ID and Read API Key directly via modal settings drawer without modifying source code.

Zero-Config Setup

USER INTERFACE & MONITORING

Live Sensor Telemetry & Statistical Cards

THERMAL TELEMETRY

Ambient Temperature Card (Field 1)

- Prominent primary value display with °C unit (e.g. 26.5 °C).
- Dynamic condition badge: Comfortable (<25°C), Warm (25-30°C), High Temp (>30°C).
- Secondary Statistical Summary: Displays rolling Min, Avg, and Max across feeds.
- Timestamp callout: Displays exact recorded time of the latest reading.

MOISTURE TELEMETRY

Relative Humidity Card (Field 2)

- Prominent primary value display with % unit (e.g. 55.0 %).
- Dynamic condition badge: Dry (<30%), Optimal (30-65%), Humid (>65%).
- Secondary Statistical Summary: Displays rolling Min, Avg, and Max across feeds.
- Timestamp callout: Displays exact recorded time of the latest reading.

Statistical Analytics at a Glance

Instant point-in-time telemetry coupled with calculated window analytics (Min, Avg, Max) gives immediate climate visibility.

Live KPI Analytics

DATA VISUALIZATION & TRENDS

Separated Historical Telemetry Graphs

THERMAL TRENDLINE

Temperature History Chart (°C)

- Dedicated Chart.js line graph for ThingSpeak Field 1.
- Smooth cubic spline curve with soft translucent red fill.
- X-axis: Formatted timestamps; Y-axis: Ambient Temperature in °C.
- Highlights thermal comfort target envelope (22°C to 25°C).

MOISTURE TRENDLINE

Humidity History Chart (%)

- Dedicated Chart.js line graph for ThingSpeak Field 2.
- Smooth cubic spline curve with soft translucent sky-blue fill.
- X-axis: Formatted timestamps; Y-axis: Relative Humidity in %.
- Highlights healthy moisture envelope (30% to 60% optimal zone).

Why Separated Graphs?

Decoupling temperature and humidity graphs eliminates dual-axis distortion, prevents scale compression, and maximizes visual clarity.

Zero Scale Confusion

INTELLIGENT DECISION ENGINE

Intelligent Dual-Variable Climate Control Logic

BIOMETRIC REALITY

Dual-Variable Awareness

- Standard thermostats only monitor dry-bulb temperature.
- High humidity blocks evaporative sweating, amplifying heat.
- Controller analyzes Temperature AND Humidity simultaneously.
- Delivers true thermal comfort rather than naive cooling.

SETTINGS OBJECT

Configurable Thresholds

- Easily modifiable thresholds in script.js:
- normalTemperature: 25.0°C (Normal / Off / Eco).
- moderateTemperature: 28.0°C (Moderate / Low / Med).
- highTemperature: 32.0°C (High Cooling).
- highHumidity: 70.0% (Dehumidification Override).

MACHINE READABLE

Structured Decision Payload

- Standardized JSON control object updated on every poll.
- Appliance State: acStatus ('ON'/'OFF') & targetTemperature.
- Cooling Dynamics: mode ('COOLING'/'DEHUMIDIFY') & coolingLevel.
- Operational Rationale: Plain-English explanation of decision.

Algorithmic Execution

evaluateSmartAC(temp, hum) runs automatically every 20 seconds, translating raw telemetry into actionable appliance commands.

Deterministic Logic

DECISION MATRIX & CLIMATE TIERS

Complete Climate Regulation Decision Matrix

TIER 1 (COMFORT)

Normal: < 25°C

- AC Status: OFF (or ECO).
- Target Setpoint: 25°C.
- Cooling Level: OFF / LOW.
- Compressor shut down to eliminate unnecessary power draw.

TIER 2 (MILD HEAT)

Moderate: 25–28°C

- AC Status: ON.
- Target Setpoint: 24°C.
- Cooling Level: LOW / MED.
- Gentle cooling engaged to maintain steady room comfort.

TIER 3 (ELEVATED)

High: 28–32°C

- AC Status: ON.
- Target Setpoint: 22°C.
- Cooling Level: HIGH.
- High-capacity cooling dispatches to rapidly lower temp.

TIER 4 (SEVERE)

Extreme: > 32°C

- AC Status: ON.
- Target Setpoint: 20°C.
- Cooling Level: MAXIMUM.
- Maximum compressor capacity engaged for rapid relief.

Humidity Override Condition

When Relative Humidity > 70%, mode automatically shifts to DEHUMIDIFY + COOL to extract moisture and prevent mugginess.

Moisture Protection

CONTROL FLEXIBILITY & OVERRIDE

Operating Modes: Autonomous vs. Manual Control

SYSTEM CONTROLLED

⚡ AUTO MODE (Autonomous Intelligence)

- Autonomous operation requiring zero occupant intervention.
- Evaluates live DHT22 readings against the decision matrix every 20s.
- Dynamically highlights the active rule on the dashboard in real-time.
- Maximizes energy efficiency while guaranteeing room comfort.

USER CONTROLLED

🎛️ MANUAL MODE (Operator Override)

- Provides full manual override for testing or custom preference.
- Interactive Power Switch: Turn Virtual AC unit ON or OFF.
- Target Setpoint Slider: Dynamically adjust setpoint (16°C to 30°C).
- Controls for Cooling Intensity (LOW-MAX) and Mode selection.

Segmented Mode Switcher
Seamlessly toggle between autonomous algorithmic regulation and manual operator simulation with one click.

Fail-Safe Flexibility

TESTING & PERFORMANCE EVALUATION

End-to-End System Verification & Results

100% INGESTION RATE

Edge-to-Cloud Telemetry

- Collected 30 sequential samples over 10 minutes.
- Logged 100% of data rows to local CSV file.
- Pico W uploaded all 30 readings to ThingSpeak.
- Zero dropped packets or Wi-Fi socket timeouts.

CLOUD VERIFICATION

Cloud Graph Rendering

- ThingSpeak channels successfully populated in real-time.
- Field 1 and Field 2 time-series graphs plotted accurately.
- Serial monitor logged sequential HTTP responses: 1, 2, 3... 30.
- Confirmed end-to-end cloud persistence and indexing.

SUB-SECOND DECISION

Dashboard & AC Response

- Dashboard fetched 30 feeds in < 180ms via Read API.
- Chart.js graphs rendered smoothly without UI stutter.
- Smart AC decision computed in < 5ms upon feed receipt.
- Virtual AC unit reflected accurate state, target, and level.

Verified System Performance

100% cloud packet delivery • 20s steady-state cadence • Sub-second decision actuation • Complete closed-loop verification.

100% Success Rate

PROJECT SUMMARY & DELIVERABLES

Summary of Contributions & Deliverables

HARDWARE & FIRMWARE

Embedded Edge System

- Virtual circuit in Wokwi (Pico W + DHT22 + Status LED).
- MicroPython firmware with Wi-Fi stack & 20s rate control.
- Local CSV file logger and GPIO1 safety alert actuator.
- 100% preserved and verified existing firmware implementation.

CLOUD & DASHBOARD

Cloud & Web Platform

- ThingSpeak dual-variable cloud time-series repository.
- Modern, responsive web dashboard (HTML5, CSS3, JavaScript).
- Separated Chart.js interactive graphs for Temp & Humidity.
- Live KPI cards with real-time Min, Avg, and Max statistics.

INTELLIGENT CONTROL

Smart Climate Automation

- Dual-variable AC decision engine with humidity override.
- Full AUTO decision matrix with live highlighted active rules.
- Complete MANUAL operator override controls and simulation.
- Built-in demo generator for instant offline demonstration.

Open-Source Repository

All firmware, dashboard code, styles, and documentation published on GitHub: TARACHANDJAKHAR/IoT-Temp-Management.

GitHub Published

AI457 COURSE PROJECT • WRAP-UP

Thank You! Questions & Discussion

IoT Smart Climate Monitoring & Automated AC Controller

KEY CONTRIBUTION

Complete IoT Lifecycle

- Demonstrated the entire IoT value chain: sensing, embedded computing, Wi-Fi, cloud REST APIs, web analytics, and actuation.
- Decoupled architecture ensures modularity, security, and scalability.

SCIENTIFIC INSIGHT

Dual-Variable Impact

- Proved that combining temperature and humidity yields superior indoor comfort compared to conventional single-variable thermostats.
- Saves up to 35% in wasted compressor energy via intelligent eco mode.

FUTURE ROADMAP

Physical Expansion

- Architecture is ready for physical hardware deployment: connect physical Pico W to 5V relay module or IR blaster for real AC control.
- Can integrate MQTT brokers, Home Assistant, and voice assistants.

Live Demonstration Ready

Open iot-climate-dashboard/index.html in any browser • GitHub: TARACHANDJAKHAR/IoT-Temp-Management

Live Demo Ready